

D116

Advanced Pharmacology for the Advanced Practice Nurse

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A patient stable on 100 mg IV of a drug is switching to oral therapy before discharge from a large academic hospital. The drug's oral bioavailability is 50%. The patient prefers once-daily dosing and lives with her daughter. What oral daily dose gives the same exposure?

- A. 150 mg
- B. 200 mg
- C. 50 mg
- D. 100 mg

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A patient starts a scheduled drug with a half-life of 8 hours. He is a night-shift security guard and takes it with meals. About when will levels reach steady state?

- A. 16 hours.
- B. Faster if the dose is doubled.
- C. About 32–40 hours.
- D. 8 hours.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A drug reaches steady state after about 5 days of regular dosing. What is its approximate half-life?

- A. About 1 day.
- B. 25 days.
- C. 12 hours.
- D. 5 days.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A patient on stable warfarin with an INR of 2.5, who walks her dog daily, is prescribed fluconazole, a CYP2C9 inhibitor, for a fungal infection. She keeps a medication log. What change should the prescriber expect and plan to monitor?

- A. No change, because antifungals never interact.
- B. A rise in INR and bleeding risk; monitor INR closely and adjust.
- C. No change, because she keeps a medication log.
- D. A fall in INR.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A patient on an oral contraceptive starts rifampin, a strong CYP3A4 inducer. What is the main concern?

- A. Reduced contraceptive levels and possible failure; advise backup contraception.
- B. No interaction, because rifampin is an antibiotic.
- C. Higher contraceptive levels and clot risk.
- D. Rifampin levels rise from the contraceptive.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A patient on lithium for bipolar disorder, who works as a landscaper in a hot climate, starts to feel thirsty and has a coarse tremor. He enjoys cooking. What is the best next step?

- A. Wait for his next routine visit.
- B. Check a serum lithium level and assess fluid status and kidney function.
- C. Stop all fluids to reduce thirst.
- D. Increase the dose, since tremor means undertreatment.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A clinician needs a loading dose for an 80-kg patient. The target plasma concentration is 15 mg/L and the volume of distribution is 0.5 L/kg. The drug's half-life is 6 hours and it is made by a well-known manufacturer. What loading dose is needed?

- A. 2.7 mg
- B. 90 mg
- C. 600 mg
- D. 1,200 mg

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

8. [TRANSFER — reversed stem / different variable]

A 700-mg loading dose is given to a patient with a volume of distribution of 35 L. What initial plasma concentration is expected?

- A. 0.05 mg/L
- B. 24,500 mg/L
- C. 35 mg/L
- D. 20 mg/L

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

9.

A 70-year-old man who plays golf weekly weighs 72 kg and has a serum creatinine of 1.4 mg/dL. He takes a renally cleared drug. His wife manages his pill box. What is his estimated creatinine clearance by Cockcroft-Gault?

- A. 42.5 mL/min
- B. 70 mL/min
- C. 50 mL/min
- D. 98 mL/min

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

10. [TRANSFER — reversed stem / different variable]

An 80-year-old woman weighs 60 kg with a serum creatinine of 1.0 mg/dL. What is her estimated creatinine clearance?

- A.** 50 mL/min
- B.** 58.8 mL/min
- C.** 100 mL/min, because the creatinine is normal.
- D.** 42.5 mL/min

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

An 82-year-old retired pharmacist who lives in a two-story home asks for diphenhydramine to help him sleep. He has an enlarged prostate. What is the best response?

- A.** Approve it, since he was a pharmacist.
- B.** Approve it, since it is sold over the counter.
- C.** Double the dose if he still cannot sleep.
- D.** Advise against it due to anticholinergic effects (confusion, falls, urinary retention) and discuss safer options.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

Drug A lowers blood pressure by 10 mmHg at 5 mg and never more than 15 mmHg at any dose. Drug B needs 50 mg to lower it by 10 mmHg but can lower it by 30 mmHg at higher doses. Drug A is newer and heavily advertised. Which statement is correct?

- A.** Drug A is more potent; Drug B is more efficacious.
- B.** Drug A is better because it is newer.
- C.** Drug B is more potent because it lowers pressure more.
- D.** Drug A is more efficacious because it works at a lower dose.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A patient stable on 100 mg IV of a drug is switching to oral therapy before discharge from a large academic hospital. The drug's oral bioavailability is 50%. ~~The patient prefers once daily dosing and lives with her daughter.~~ What oral daily dose gives the same exposure?

Target word: oral dose **Decides:** Oral = IV / F.

A	Confusion set	Adds 50% instead of dividing by 0.5.
B	Correct	$100 / 0.5 = 200$ mg.
C	Inverted operation	Multiplies by F instead of dividing.
D	Dropped element	Ignores bioavailability.

2. Answer C Concept 2

A patient starts a scheduled drug with a half-life of 8 hours. ~~He is a night shift security guard and takes it with meals.~~ About when will levels reach steady state?

Target word: time to steady state **Decides:** 4-5 half-lives.

A	Dropped element	Stops at two half-lives.
B	Inverted operation	Dose size does not change time to steady state.
C	Correct	$4-5 \times 8$ hours.
D	Wrong stem number	One half-life.

3. Answer A Concept 2 · transfer

A drug reaches steady state after about 5 days of regular dosing. What is its approximate half-life?

Target word: solve for half-life **Decides:** 5 half-lives $\approx$ 5 days.

A	Correct	5 days / 5 half-lives.
B	Inverted operation	Multiplies instead of divides.
C	Confusion set	Assumes 10 half-lives are needed.
D	Wrong stem number	The time to steady state, not the half-life.

4. Answer B Concept 3

A patient on stable warfarin with an INR of 2.5, ~~who walks her dog daily,~~ is prescribed fluconazole, a CYP2C9 inhibitor, for a fungal infection. ~~She keeps a medication log.~~ What change should the prescriber expect and plan to monitor?

Target word: interaction effect **Decides:** Inhibitor raises warfarin levels.

A	Absolute language	Azole antifungals are well-known inhibitors.
B	Correct	Inhibiting warfarin metabolism raises its effect.
C	Camouflage	Logging does not prevent the interaction.
D	Inverted operation	That would follow an inducer.

5. Answer A Concept 3 • transfer

A patient on an oral contraceptive starts rifampin, a strong CYP3A4 inducer. What is the main concern?

Target word: inducer effect **Decides:** Inducer lowers levels of affected drugs.

A	Correct	Induction speeds metabolism of the contraceptive.
B	Absolute language	Drug class does not rule out enzyme effects.
C	Inverted operation	That would follow an inhibitor.
D	Confusion set	Reverses which drug is affected.

6. Answer B Concept 4

A patient on lithium for bipolar disorder, who works as a landscaper in a hot climate, starts to feel thirsty and has a coarse tremor. ~~He enjoys cooking.~~ What is the best next step?

Target word: monitoring priority **Decides:** Narrow index drug with signs of toxicity; heat and fluid loss raise levels.

A	Dropped element	Drops urgent monitoring.
B	Correct	Narrow therapeutic index drugs need level checks when toxicity is possible.
C	Adjacent category	Dehydration raises lithium levels.
D	Inverted operation	Coarse tremor suggests toxicity, not low levels.

7. Answer C Concept 5

A clinician needs a loading dose for an 80-kg patient. The target plasma concentration is 15 mg/L and the volume of distribution is 0.5 L/kg. ~~The drug's half-life is 6 hours and it is made by a well-known manufacturer.~~ What loading dose is needed?

Target word: loading dose **Decides:** Target $\times$ Vd = 15 $\times$ 40 L.

A	Inverted operation	Divides Vd by target instead of multiplying.
B	Camouflage	Multiplies target by half-life (15 $\times$ 6), which does not set a loading dose.
C	Correct	0.5 L/kg $\times$ 80 kg = 40 L; 15 mg/L $\times$ 40 L = 600 mg.
D	Dropped element	Uses 80 L, dropping the 0.5 L/kg factor.

8. Answer D Concept 5 • transfer

A 700-mg loading dose is given to a patient with a volume of distribution of 35 L. What initial plasma concentration is expected?

Target word: solve for concentration **Decides:** C = dose / Vd.

A	Confusion set	Divides Vd by dose.
B	Inverted operation	Multiplies dose by Vd.
C	Wrong stem number	The Vd, not the concentration.
D	Correct	700 / 35 = 20.

9. Answer C Concept 6

A 70-year-old man ~~who plays golf weekly~~ weighs 72 kg and has a serum creatinine of 1.4 mg/dL. He takes a renally cleared drug. ~~His wife manages his pill box.~~ What is his estimated creatinine clearance by Cockcroft-Gault?

Target word: CrCl **Decides:** $(140 - 70) \times 72 / (72 \times 1.4)$.

A	Adjacent category	Applies the 0.85 female factor to a man.
B	Dropped element	Drops the creatinine term.
C	Correct	$70 \times 72 / 100.8 = 50$.
D	Inverted operation	Multiplies by SCr instead of dividing (70×1.4) .

10. Answer D Concept 6 · transfer

An 80-year-old woman weighs 60 kg with a serum creatinine of 1.0 mg/dL. What is her estimated creatinine clearance?

Target word: CrCl for a woman **Decides:** $(140 - 80) \times 60 / 72 \times 0.85$.

A	Dropped element	Drops the 0.85 female factor.
B	Inverted operation	Divides by 0.85 instead of multiplying.
C	Absolute language	A normal SCr can hide reduced clearance in small older adults.
D	Correct	$60 \times 60 / 72 = 50$; $\times 0.85 = 42.5$.

11. Answer D Concept 7

An 82-year-old retired pharmacist ~~who lives in a two-story home~~ asks for diphenhydramine to help him sleep. He has an enlarged prostate. What is the best response?

Target word: Beers guidance **Decides:** First-generation antihistamine in an older adult.

A	Camouflage	His career does not change the drug's risks.
B	Absolute language	OTC status does not mean safe for older adults.
C	Inverted operation	Increases anticholinergic harm.
D	Correct	Beers Criteria list first-generation antihistamines to avoid.

12. Answer A Concept 8

Drug A lowers blood pressure by 10 mmHg at 5 mg and never more than 15 mmHg at any dose. Drug B needs 50 mg to lower it by 10 mmHg but can lower it by 30 mmHg at higher doses. ~~Drug A is newer and heavily advertised.~~ Which statement is correct?

Target word: potency vs. efficacy **Decides:** A needs less drug; B has the higher maximum effect.

A	Correct	Potency = dose needed; efficacy = maximum effect.
B	Camouflage	Newness does not decide potency or efficacy.
C	Inverted operation	Reverses the two terms.
D	Confusion set	Confuses potency with efficacy.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	2	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	12	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Bioavailability and first-pass effect	Oral drugs pass through the gut and liver first, so only a fraction (F) reaches circulation; oral dose = IV dose / F to match exposure.	Converting IV to oral mg-for-mg.	An IV dose and an oral bioavailability percentage.
2	Half-life and steady state	With regular dosing, steady state is reached in about 4–5 half-lives, regardless of dose size.	Thinking a bigger dose reaches steady state faster.	A half-life and a question about when levels stabilize.
3	CYP450 inhibition vs. induction	An enzyme inhibitor raises levels of drugs metabolized by that enzyme (toxicity risk); an inducer lowers them (loss of effect), often over days to weeks.	Reversing the direction of inhibitors and inducers.	A new drug added to a stable regimen with a narrow-margin drug.
4	Narrow therapeutic index	When toxic and effective levels are close, dosing needs serum-level or lab monitoring and careful changes.	Treating every drug as safe to titrate by symptoms alone.	Drugs such as lithium, digoxin, warfarin, or phenytoin.
5	Loading dose	Loading dose = target concentration × volume of distribution; it depends on Vd, not on half-life or clearance.	Using clearance or half-life to size a loading dose.	A target concentration and a Vd.
6	Renal dose adjustment	Estimate creatinine clearance (Cockcroft–Gault: $(140 - \text{age}) \times \text{weight} / (72 \times \text{SCr})$, × 0.85 if female) and adjust renally cleared drugs.	Relying on a normal-looking serum creatinine in a small, older adult.	Age, weight, sex, and serum creatinine given for a renally cleared drug.
7	Beers Criteria	Drugs with strong anticholinergic or sedative effects (e.g., first-generation antihistamines) are generally avoided in older adults due to falls, confusion, and retention.	Assuming an over-the-counter drug is safe for older adults.	An older adult asking about a sleep aid or allergy medication.
8	Potency vs. efficacy	Potency is how much drug is needed for an effect; efficacy is the maximum effect a drug can produce.	Calling the drug with the smaller mg dose the stronger or more effective drug.	Two drugs compared by dose size vs. by maximum response.

Course scope (public description): Advanced Pharmacology for the Advanced Practice Nurse prepares the graduate nursing student for the role of an advanced practice nurse with the competencies and skills for prescribing and monitoring medication safely and effectively. This course will prepare the graduate nursing student to apply pharmacotherapeutics in primary care settings by utilizing the pivotal basis of pharmacokinetics and pharmacodynamics. This course will also prepare the graduate nursing student to select the correct medication, describe the rationale for that selection to the patient, family, and other providers, and to effectively monitor the patient to promote positive drug outcomes.